

# Data Flow Diagram

|               |                                                    |
|---------------|----------------------------------------------------|
| Date          | 04 July 2026                                       |
| Team ID       | SOLO-CCFD-2026                                     |
| Project Name  | Credit Card Fraud Detection using Machine Learning |
| Maximum Marks | 2 Marks                                            |

**Data Flow Diagram — Description** (see legend below; a visual diagram image should accompany this text in the final submission).

| Symbol              | Name            | Description                                                                                                |
|---------------------|-----------------|------------------------------------------------------------------------------------------------------------|
| Oval                | External Entity | Cardholder, Payment Gateway, Compliance Officer                                                            |
| Rectangle w/ number | Process         | 1. Capture Transaction, 2. Preprocess Features, 3. Score with ML Model, 4. Flag/Approve, 5. Notify Analyst |
| Solid rectangle     | Data Store      | Transaction Log DB, Trained Model Store (.pkl), Alert/Case DB                                              |
| Labeled arrow       | Data Flow       | Raw transaction → preprocessed features → fraud probability → decision → alert/report                      |

**Flow narrative:** (1) Payment Gateway sends raw transaction (Time, Amount, V1–V28) to the Preprocessing process. (2) Preprocessing scales Time/Amount and forwards the feature vector to the ML Scoring process, which loads the trained model from the Model Store. (3) The Scoring process outputs a fraud probability to the Decision process, which either auto-approves or raises a flag. (4) Flagged transactions are written to the Alert/Case DB and pushed to the Analyst Dashboard for the Compliance Officer to review. (5) All transactions (flagged or not) are logged to the Transaction Log DB for audit and future retraining.